

THE EQUITY EXPRESSION OF THE COPPER SMELTING BOTTLENECK THESIS

The Fee That Went Negative

Copper has spent August at or near the highest prices ever recorded. The obvious inference is that everyone who touches the metal is making money. That inference is wrong, and the way it is wrong is the most useful thing in the market this month. The fee a smelter charges to turn copper ore into copper metal — the treatment charge — settled at zero for 2026 and has been negative in the spot market since 2024. Smelters are now paying miners for the privilege of processing their ore.

A boiler leak on 8 August at the Gresik smelter in Indonesia, which processes ore from the Grasberg mine, tightened the refined market further and pushed prices back to within half a cent of the record. The question worth asking is not whether copper is scarce. It is which kind of copper is scarce, and who is paid when it is.

A reframe worth holding onto

The copper market is not one market. It is two, joined by a furnace. Mined copper arrives as concentrate — rock that is roughly a quarter copper. Refined copper leaves the smelter as cathode, which is what the exchange price refers to.

For a decade the constraint sat on the smelting side, so smelters were paid well to convert. That has inverted. There is now too much furnace capacity chasing too little ore, and the conversion fee has collapsed through zero. The scarce thing is not metal. It is rock.

Method. Every external figure in this edition carries a source and an as-of date, and comes from company disclosure, an official agency, or dated market reporting. Prices are as at the dates stated and were not re-checked at the time of writing. **The internal leg of this edition is missing.** The Heyokha research Wiki could not be reached on this run — the connector was not authorised — so no internal view, no named research provider and no Wiki page underpins anything below. Every claim here rests on external reporting alone. Where this edition would normally set an internal house view against outside evidence, it instead sets external sources against each other. This is a material degradation and is described in full in the closing notes.

Picture a farming valley with one wheat harvest and five flour mills. The mills do not own wheat; they are paid a fee per sack to grind it. When mills were few the fee was generous, so over twenty years four more mills were built on the promise of a growing harvest that never quite arrived. Now five sets of grinding stones bid for one harvest. The fee falls to nothing, then below it: a miller offers to grind for free, then offers the farmer a payment on top, simply to keep his stones turning and his workers employed. Meanwhile the price of bread is the highest in living memory. The farmer is rich. The baker charges more. And the miller — the one person who physically makes the flour — is paying to work. He survives because he also sells the bran and the husk, and this year bran happens to be worth a fortune. That is the copper market in August 2026, and the bran matters more than anything else here.

THE STORY IN 60 SECONDS

\$0/t

The 2026 annual benchmark treatment charge, settled in January 2026, down from about \$21 per tonne for 2025 and \$80 per tonne for 2024. Spot rates have been negative since 2024 and at all-time lows. *IEA commentary on copper smelting; benchmark terms reported 19 December 2025.*

\$6.60/lb

Copper, 14 August 2026 — about \$14,550 a tonne. Up 4.86% on the month and 47.60% over twelve months. The COMEX record of \$6.7045/lb was set on 5 August 2026. *Trading Economics, as at 14 August 2026.*

900,000t

Copper redirected into United States warehouses during 2026 ahead of a possible import tariff, against LME warehouse stocks of 218,300 tonnes on 11 August. *Reported June–August 2026, citing Goldman Sachs estimates.*

WHAT ACTUALLY HAPPENED

Two accidents, one furnace, and a record that would not quite break

On 8 August 2026 a boiler leak occurred at the Gresik smelter in East Java, operated by PT Smelting, a joint venture between Mitsubishi Materials and PT Freeport Indonesia. The plant was shut for assessment and repair with no restart date decided. Gresik processes concentrate from Grasberg, one of the largest copper and gold deposits in the world. Freeport's newer Manyar smelter, on the same island, was due to resume production in September; together the two facilities target close to 400,000 tonnes of cathode output a year.

The market reaction was immediate and specific. On 11 August the COMEX September contract touched \$6.7005 per pound, equivalent to \$14,772 a tonne, leaving it four-tenths of a cent below the record of \$6.7045 set on 5 August. December traded as high as \$6.7980, and 2027 contracts changed hands above \$7. On the London Metal Exchange three-month copper was \$14,195.50 a tonne, and — the detail that matters — cash copper commanded a premium of \$138 over the three-month contract, the widest since October. A cash premium of that size is the physical market saying it wants metal now, not later. LME warehouse stocks fell 4,675 tonnes to 218,300 tonnes on the same day.

This was the second blow to the same ore body in under a year. On 24 September 2025 a rush of some 800,000 tonnes of wet material overwhelmed the Grasberg Block Cave operation. Two employees were killed and five were reported missing. Freeport declared force majeure. Goldman Sachs estimated the incident would remove 250,000 to 260,000 tonnes of annual copper production across 2025 and 2026. Freeport's second-quarter 2026 results, reported on 23 July, showed the mine running at roughly half rates, with guidance to reach about 65% in the second half of 2026, 80% by mid-2027, and close to full capacity only by the end of 2027.

Grasberg is the vivid case, but it is not an isolated one. Codelco, the Chilean state producer and the largest single source of mined copper on earth, cut its 2026 guidance to about 1.34 million tonnes from roughly 1.37 million, and its chairman abandoned the medium-term target of 1.7 million tonnes a year outright in an interview on 28 July 2026, saying that for seven years the company had not met its projections. Chile's copper commission Cochilco cut the national 2026 forecast to 5.27 million tonnes, down 2.6% year on year. Antofagasta, reporting first-half results on 13 August, cut full-year guidance to 625,000–655,000 tonnes from 650,000–700,000 after severe weather shut Los Pelambres. The pattern is consistent: the world keeps finding reasons not to produce the ore it has already promised.

THE ESSENTIAL DISTINCTION

The exchange price of copper is the miner's revenue. It is not the smelter's. A smelter is paid to *convert* metal, not to own it — so a record copper price passes straight through its accounts and out the other side. When ore is scarce relative to furnace capacity, the smelter's fee falls even as the metal's price rises. The two can move in opposite directions indefinitely, and in 2026 they have.

THE MECHANISM

One tonne of concentrate, followed through the chain

1 · The mine sells rock

A tonne of copper concentrate is roughly a quarter to a third copper by weight, plus gold, silver and sulphur. The miner is paid the exchange copper price on the contained metal, less deductions — the largest of which is the treatment charge.

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2 · The fee inverts

Historically the smelter deducted a fee to convert. The 2024 benchmark was \$80 a tonne; 2025 was \$21.25; 2026 settled at zero, and spot has been below zero since 2024. The miner now keeps effectively the whole copper price.

→

3 · The furnace earns elsewhere

Standard contracts pay the miner for 96.5% of contained copper. Anything the smelter recovers above that is free metal. It also keeps the sulphur as sulphuric acid, plus gold and silver — and sells surplus heat.

→

4 · Cathode meets the price

Refined cathode is sold at the exchange price plus a regional physical premium. That premium — not the fee — is where location scarcity is expressed, and it is where the tariff distortion currently lives.

Mechanism as described in Fastmarkets analysis of negative concentrate treatment charges; benchmark levels per IEA commentary and reporting of the 2025 and 2026 annual settlements (19 December 2025). Payability of 96.5% is the standard contractual convention, not a figure specific to any named producer.

Step three is the whole edition. A smelter that loses money on its stated business can remain not merely solvent but comfortably profitable if the residues are valuable enough — and in 2026 they are extraordinarily valuable. Jiangxi Copper, China's largest smelter, disclosed that sales of sulphuric acid accounted for 14.65% of its total gross profit in 2025, more than the combined gross profit from its finished copper products. The by-product is worth more than the product.

Aurubis, Europe's largest copper smelter, reported the same thing from the other end of the world. In results published on 6 August 2026, operating earnings before taxes for the first nine months of its financial year reached €374m, up 31% on the prior year's €286m, with the third quarter alone at €149m against €57m a year earlier. The company moved its full-year guidance of €425–525m to the upper end of the range. It named significantly higher sulphuric acid revenues as a key driver, alongside higher precious metal prices and recycling revenues — while noting that spot treatment charges for concentrates continued to decline. A smelter posting its best earnings in years, explicitly not because of smelting.

\$80 → \$0

The annual benchmark treatment charge per tonne of concentrate, 2024 to 2026. The 2025 settlement was \$21.25. The fee the conversion industry lives on has been extinguished in two years.

14.65%

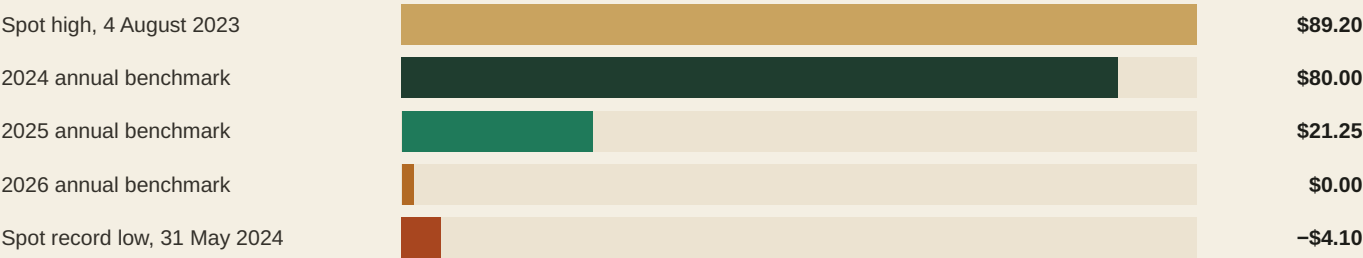
Sulphuric acid's share of Jiangxi Copper's total gross profit in 2025 — more than its finished copper products contributed. The residue outearned the metal.

~50%

China's share of global copper smelting capacity, having accounted for over 90% of the growth in world smelter output since 2005. The overcapacity has one principal address.

THE CENTRAL COMPARISON

What a tonne of concentrate paid the smelter to process it



Bars are illustrative and scaled to the highest positive value shown; the negative bar is drawn as a magnitude, not a direction. Spot figures from Fastmarkets assessments of copper concentrate treatment charges; annual benchmarks per IEA commentary and reporting of the 2025 and 2026 settlements. Treatment charges are quoted per dry metric tonne of concentrate and are accompanied by a separate refining charge in cents per pound, which has followed the same path to zero.

The obvious conclusion from that chart is that smelters must close, and that when they do, concentrate will free up, the fee will recover, and refined output will fall in the meantime. That is the textbook adjustment, and it is the argument on which a good deal of bullish copper positioning rests. It has not happened. In late 2025 the China Smelter Purchase Team, a group of sixteen major Chinese smelters, agreed to cut production by more than 10% in 2026 to counter the distortion. By April 2026, three of the largest members had not reduced their output guidance at all. Jiangxi Copper raised its 2026 target to 2.39 million tonnes from 2.38 million. Yunnan Copper raised its to 1.71 million tonnes from 1.64 million. Only Daye Nonferrous edged down, to 713,000 tonnes from 716,000. The same group has now declined to set quarterly treatment charge guidance for the fifth consecutive time.

The reason the cartel does not hold is the reason the analogy ends with bran. A smelter whose acid, gold and silver revenues are booming has no commercial motive to idle a furnace, and a Chinese smelter with local employment obligations has still less. The Chinese authorities have halted around 2 million tonnes of planned new smelting capacity, which addresses the future problem. It does nothing about the furnaces already built.

A USEFUL MENTAL MODEL

A shortage in one warehouse is not a shortage in the world. Roughly 900,000 tonnes of copper have been redirected into United States warehouses during 2026 in anticipation of an import tariff. That metal did not stop existing; it changed address. The tightness visible in London is partly the accounting shadow of abundance in Baltimore — which means part of the price signal is a policy artefact, and will unwind when the policy resolves.

THE STATE OF PLAY

What has actually shifted

WHAT HAS CHANGED	WHAT IT LOOKS LIKE IN THE DATA	WHY IT CUTS BOTH WAYS
The conversion fee has been extinguished	Annual benchmark from \$80/t (2024) to \$21.25/t (2025) to \$0/t (2026); spot negative since 2024 and at record lows	It transfers economics decisively from smelters to miners — but a fee at zero cannot fall much further, so it has stopped being a source of incremental change

WHAT HAS CHANGED	WHAT IT LOOKS LIKE IN THE DATA	WHY IT CUTS BOTH WAYS
Mine supply keeps disappointing	Codelco's 1.7Mt target abandoned 28 July; Chile's 2026 forecast cut to 5.27Mt; Antofagasta guidance cut 13 August; Grasberg at roughly half rates	It is the strongest support under the price — and it is also the most reliably mean-reverting variable in mining, since deferred tonnes are usually delayed rather than destroyed
By-products have become the earnings engine	Sulphuric acid 14.65% of Jiangxi Copper's 2025 gross profit; Aurubis names acid revenues as a key driver of a 31% rise in nine-month operating EBT	It keeps loss-making furnaces alive, delaying the supply adjustment bulls need — and it makes smelter earnings a bet on acid and gold rather than on copper
Metal has relocated rather than been consumed	About 900,000t redirected to US warehouses in 2026; US inventories projected near 1.8Mt by year end; LME stocks 218,300t on 11 August	Ex-US tightness is real for anyone who needs metal outside America — but it is positional, and a tariff decision in either direction releases or strands it
High prices are starting to bite demand	Copper eased to near two-week lows in mid-August on weaker Chinese buying; the Yangshan import premium fell to \$96/t from \$115/t the previous month	It is the mechanism that eventually caps any commodity spike — but premiums are noisy and a single month is not a trend

THE ARGUMENT

Bull and bear, stated fairly

The case for higher

- **Ore is the binding constraint and it is not responding.** Codelco abandoned its 1.7Mt medium-term goal on 28 July after seven straight years of missed projections; Cochilco cut Chile's 2026 forecast to 5.27Mt; Antofagasta cut guidance on 13 August. Three downgrades in three weeks.
- **The geology is genuinely deteriorating.** The IEA notes ore grades have fallen roughly 40% since 1991, and that only 5% of the past 35 years' discoveries came in the last decade. Lower grade means more rock and more capital per tonne of metal.
- **Negative treatment charges are the market's own scarcity signal.** Smelters bidding above the full value of contained metal is not sentiment; it is operators with furnaces to fill revealing what concentrate is worth to them.
- **Disruption is compounding, not isolated.** The Grasberg mud rush removed an estimated 250,000–260,000t of annual output across 2025–26; the 8 August Gresik leak then hit the refining of what ore remains.
- **The curve prices scarcity years out.** COMEX December traded at \$6.7980 and 2027 contracts above \$7 on 11 August — not merely a spot squeeze.
- **Miner economics are already validating the price.** Antofagasta's first-half EBITDA rose 27% to \$2,840.5m on 9% lower volumes, at a 63.4% margin. Glencore's first-half adjusted EBITDA reached \$10.1bn.

The case for lower

- **The tightness is partly an address change.** Around 900,000t moved into US warehouses in 2026 ahead of a tariff, with US inventories projected near 1.8Mt by year end. Goldman Sachs was reported to have moved from a 490,000t global surplus in April to a 640,000t ex-US deficit by June — the same metal, re-sorted.
- **The tariff that caused it has not arrived.** Commerce missed a 30 June 2026 deadline on refined copper; the reported proposal is 15% from January 2027, rising to 30% in 2028. Delay or dilution releases that stockpile.
- **Reputable forecasts sit far below the tape.** Analysis published 9 July 2026 cited Goldman Sachs fair value near \$11,500/t and a fourth-quarter forecast of \$11,200/t, against roughly \$14,550 on 14 August — plus a 300kt global surplus for 2026.
- **Demand is visibly flinching.** Copper slipped to near two-week lows in mid-August as high prices weakened Chinese buying, and the Yangshan premium fell to \$96/t from \$115/t. Fabricators are reported to be reviewing aluminium substitution in non-critical wiring.
- **The supply adjustment bulls expect is not happening.** Chinese smelters pledged cuts above 10% for 2026; by April the three largest had raised or held guidance. Refined output is not falling, whatever the fee says.
- **High prices summon scrap.** Elevated prices accelerate secondary supply from recycling and dormant stockpiles — a source that responds in months, not the seven years a new mine takes.

An unresolved conflict

This edition would normally set the firm's internal house view against outside evidence and name the gap. That was not possible: the internal Wiki was unreachable, so there is no internal view to test. What can be shown instead is that the external record contradicts itself, sharply, and that the contradiction has not been resolved by events.

One body of reporting, citing Goldman Sachs work published around 9 July 2026, puts fundamental fair value near \$11,500 a tonne, forecasts \$11,200 for the fourth quarter, and expects a global surplus of 300,000 tonnes this year. Another, citing the same house in June, describes a 640,000-tonne deficit outside the United States and a wholesale revision from an earlier surplus view. Copper meanwhile traded around \$14,550. These are reconcilable only through geography — a world in surplus and an ex-America market in deficit, with the difference sitting in US warehouses — and if that reconciliation is right, then a large part of the record price is a tariff artefact rather than a scarcity signal. If it is wrong, one of the two forecasts is badly mistaken. There is a second, sharper contradiction inside China: sixteen smelters formally agreed to cut output by more than 10%, and then the three largest raised or maintained guidance. Both facts are documented. Only one of them can be describing the industry's actual behaviour. This edition does not resolve either conflict, and readers should be sceptical of anyone who claims to.

A smaller irony deserves recording, because it shows how tangled the chain has become. Glencore raised its 2026 copper unit cash cost guidance to 225.1 cents per pound from 203 cents in February, citing higher landed costs for diesel, sulphuric acid and sulphur. The same acid price that is rescuing smelter earnings is inflating the operating costs of the miners who leach ore. The windfall at one point in the chain is a cost at another.

THE DASHBOARD

What to watch from here

SIGNAL	PLAIN-ENGLISH READING	WHY IT MATTERS
Spot treatment charges	The fee smelters are paid, or pay, per tonne of concentrate. Negative since 2024; benchmark zero for 2026	The single cleanest read on whether ore or furnace capacity is the binding constraint. A move back above zero would say the ore squeeze is easing
Yangshan copper premium	What Chinese buyers pay above the exchange price for physical import. \$96/t in August, down from \$115/t	The most sensitive available gauge of whether Chinese demand is absorbing these prices or refusing them
LME cash-to-three-month spread	A \$138 cash premium on 11 August, the widest since October, on stocks of 218,300 tonnes	Tells you whether the shortage is immediate and physical or a forward expectation. A collapse in this spread would mark the squeeze breaking
The COMEX–LME spread	The gap between US and world copper prices, around \$400/t by early June and traded as a live gauge of tariff odds	It is the price of the policy, not the metal. It should converge towards zero if the tariff is abandoned and widen towards the tariff rate if it is confirmed
Gresik repair and Manyar restart	Gresik shut from 8 August with no restart date; Manyar was due to resume in September; together near 400,000t of annual cathode	The most concrete near-term swing factor in refined supply, and a direct test of how much of August's move was one accident
Jackson Hole, 27–29 August	The Kansas City Fed symposium, themed on financial innovation and payments. Kevin Warsh has chaired the Federal Reserve since 22 May 2026	Copper is priced in dollars and financed at the front end. A hawkish read lifts the dollar and real rates, which historically works against industrial metals regardless of the physical balance

Three postures, not one answer

Own the rock

If ore is the scarce thing, the miner captures the whole price with the treatment charge at zero. The explicit cost is that mining equities have become a bet on execution, not on copper: Codelco, Antofagasta and Freeport have all cut or abandoned targets since late July. One buys the commodity and receives operational risk alongside it.

Own the conversion

Smelters are cheap on the stated business and earning well on the residues, with acid and precious-metal credits doing the work. The explicit cost is that this is not a copper position at all. It is a position in sulphuric acid, gold and silver prices, wearing a copper label — and those can reverse without copper moving at all.

Own the dislocation

The COMEX–LME spread and the cash premium are the purest expression of the distortion, and trading houses with metal and warehousing are positioned for it. The explicit cost is that this is a policy trade with a binary resolution: a tariff decision either way compresses the spread that is the entire source of return.

The companies named below are illustrative of the mechanisms discussed in this edition. They are NOT recommendations, and no view is expressed or implied on the merits of any security. Figures are drawn from company disclosure and dated market reporting, and prices are as at the dates stated.

FCX

Freeport-McMoRan · NYSE

\$66.49 · 14 Aug 2026
Q2 results 23 Jul 2026

The purest illustration of price offsetting volume. Second-quarter revenue was \$7,029m and adjusted net income \$1.1bn, or \$0.74 a share, on copper sales of 710m pounds at a realised \$6.17/lb against \$4.54 a year earlier. Unit net cash costs were \$1.97/lb. Grasberg ran at about half rates after the September 2025 mud rush, with guidance to roughly 65% in the second half and near-full capacity only at end-2027. The 8 August Gresik leak hit the smelting of what ore it is producing.

WATCH NEXT

Whether the Grasberg ramp holds to the 65% second-half figure, and when Gresik and Manyar are both running.

ANTO

Antofagasta · London

3,585.0p · 14 Aug 2026
H1 results 13 Aug 2026

The clearest picture of what zero treatment charges do to a miner's economics. First-half EBITDA rose 27% to \$2,840.5m on revenue of \$4,479.0m, at a 63.4% margin, with pre-tax profit up 72% — achieved on copper production of 285,000 tonnes, 9% *lower* than a year earlier. Net cash costs fell 8% to \$1.22/lb. Full-year guidance was cut to 625,000–655,000 tonnes after weather shut Los Pelambres. It was also the producer that settled 2026 terms at \$0 per tonne and zero cents per pound.

WATCH NEXT

Whether the Los Pelambres restart holds the reduced guidance, and net debt, which rose to about \$4bn from \$2.8bn.

NDA

Aurubis · Frankfurt

€167.00 · 14 Aug 2026
Q3 results 6 Aug 2026

The listed test of whether a smelter can prosper while its core fee is extinguished. Nine-month operating earnings before taxes reached €374m, up 31% from €286m, with the third quarter at €149m against €57m a year earlier, and full-year guidance of €425–525m moved to the upper end. Management attributed the improvement to significantly higher sulphuric acid revenues, higher precious metal prices and recycling — while noting that spot concentrate treatment charges continued to decline. The by-products are carrying the business.

WATCH NEXT

Sulphuric acid pricing in north-west Europe, which is now the swing variable in the earnings, rather than anything copper does.

0358

Jiangxi Copper · Hong Kong

HK\$37.30 · 17 Aug 2026
2026 guidance 2.39Mt

China's largest smelter, and the company whose disclosure supplies this edition's central fact: sulphuric acid was 14.65% of total gross profit in 2025, more than finished copper products contributed. Despite belonging to the smelter group that pledged output cuts above 10% for 2026, it raised guidance to 2.39m tonnes from 2.38m. First-quarter 2026 revenue rose 25.4% to RMB139.12bn with net profit up 44.3% to RMB2.82bn. Interim results for 2026 had not been published as at 17 August.

WATCH NEXT

The interim results, and specifically whether the by-product share of gross profit has risen again from 14.65%.

GLEN

Glencore · London

548.9p · 14 Aug 2026
H1 results 5 Aug 2026

The company positioned on both sides of the dislocation. First-half adjusted EBITDA was \$10.115bn and net income \$4.405bn, with marketing adjusted EBIT of \$3.3bn, up 142%, and full-year marketing guidance of \$4.7–5.0bn. Own-sourced copper rose 15% to 397,000 tonnes. Its chief executive argued on 5 August that long-term benchmarks make less sense in a market trading increasingly on spot, adding that as both producer and trader the company can take advantage of the volatility in those differentials.

WATCH NEXT

Whether marketing earnings hold at the raised guidance once the tariff question resolves and the spread they trade compresses.

AMMN

Amman Mineral Internasional · Jakarta

Smelter certified 18 Jul 2026
Results due 27 Aug 2026

Indonesia's attempt to keep the conversion margin at home. The company's copper smelter received its completion certificate on 18 July 2026, designed for 900,000 tonnes a year of concentrate from Batu Hijau and producing cathode, gold, silver, sulphuric acid and selenium. First-quarter net sales were \$808m with EBITDA of \$508m at a 63% margin, but the smelter averaged only about 50% run rate for cathode, and the company declined to give full-year cathode guidance, prioritising stable operation. Its share price could not be verified to a consistent level across sources and is therefore omitted.

WATCH NEXT

The 27 August results, and whether smelter utilisation has moved materially above the 50% first-quarter rate.

The bottom line

What was repriced in August was not copper in general. It was the location and the form of copper, and the distinction between owning the metal and being paid to make it. The exchange price near records reflects a genuine and worsening shortage of *ore* — three separate guidance downgrades from Codelco, Cochilco and Antofagasta inside three weeks, on top of a mine that lost eight lives and a year of output — layered on top of a policy-driven relocation of roughly 900,000 tonnes into American warehouses. The treatment charge at zero is the market's own confession that furnaces are abundant and rock is not. But it also means the conversion industry has quietly stopped earning its living from copper at all, and now earns it from sulphuric acid and precious metals instead.

The evidence does not point cleanly in one direction and this edition will not pretend it does. Mine supply is genuinely failing, the forward curve prices scarcity into 2027, and miner margins are validating the price. Against that, respected forecasts published six weeks ago put fair value nearly a quarter below the current tape, Chinese buyers are visibly balking, the Yangshan premium has fallen, and the pledged Chinese smelter cuts that the bull case needs simply have not materialised. The reasonable inference is that the ore shortage is real and durable while the refined-metal shortage is substantially positional — which would mean the miners' advantage outlasts the price spike, and the spread trades do not. The next test is specific and close: whether Gresik restarts and the COMEX–LME spread narrows before Jackson Hole on 27–29 August, or after it.

Editorial note. This document is educational commentary prepared for internal use. It is not investment advice, not a recommendation, and not an offer or solicitation to buy or sell any security. Companies named are illustrative of the mechanisms discussed and no view is expressed on their merits. All external figures are attributed and carry an as-of date; prices and market data are as at the dates stated and were not re-verified at the time of publication. Internal research views, where present, are the opinions of the named research providers, are labelled as views rather than forecasts, and are not verified facts. **On this edition, no internal view is present at all — the research Wiki could not be reached, and the entire document rests on external sources.** Where sources conflict, the conflict is stated and left unresolved rather than reconciled. Figures that could not be verified have been omitted rather than estimated.

Sources and update notes

(a) Internal sources — none used. No Heyokha Wiki page underpins this edition. The Heyokha Wiki MCP connector reported that authorisation was required and could not complete an interactive sign-in from an unattended scheduled run, so *list_sources*, *list_pages*, *get_page*, *get_relations* and *search_wiki* were all unavailable. In consequence: no theme page was read, no *last_reviewed* date can be quoted, no research provider issue — 13D Research, White Box, Greed & Fear, Solid Ground, Marc Faber, HS Dent, Gary Shilling or any other — was consulted, and no internal thesis, bull case, bear case or contradictions section informed the text. Nothing in this document should be read as representing the firm's internal view, because the firm's internal view was not available to the author of this run.

(b) Limitation on the internal view. The standing purpose of this paragraph is to name the provider concentration behind the internal argument, so that a reader can discount for it. That cannot be done here. Because no Wiki page was retrieved, the usual limitation is replaced by a more serious one: this edition has no internal counterweight whatsoever, and therefore no house view against which to test external reporting. The compensating step taken was to set external sources against one another and to prefer primary disclosure — company results releases, the Federal Reserve, the IEA and the Kansas City Fed — over aggregators wherever a figure existed in both. Readers should note that several market-data figures were nonetheless sourced from commercial reporting rather than exchange primary data, that share prices sourced from quote aggregators proved unreliable during preparation, and that one price (Amman Mineral) was dropped entirely because three sources disagreed.

(c) External sources. **1.** Copper at \$6.60/lb, +4.86% on the month and +47.60% over twelve months; Yangshan premium \$96/t from \$115/t; Codelco target abandonment — Trading Economics copper commodity page, as at 14 August 2026. **2.** COMEX September \$6.7005/lb (\$14,772/t) on 11 August 2026, 0.4c below the 5 August record \$6.7045/lb; December \$6.7980/lb; 2027 contracts above \$7/lb; LME three-month \$14,195.50/t; cash premium \$138 over three-month, widest since October; LME stocks –4,675t to 218,300t; Gresik boiler leak of 8 August 2026 at PT Smelting (Mitsubishi Materials and PT Freeport Indonesia), no restart date; Manyar due to resume September; combined target near 400,000t cathode a year — MINING.COM, 11 August 2026. **3.** 2026 benchmark treatment charge \$0/t settled January 2026; 2025 benchmark about \$21/t; spot negative since 2024 at all-time lows; China about half of global smelting capacity and over 90% of growth in smelter output since 2005; Chinese smelters agreed to cut production by over 10% in 2026; around 2Mt of planned new smelting capacity halted; ore grades down about 40% since 1991; 5% of the past 35 years' discoveries made in the last decade — International Energy Agency commentary on copper smelting, 2026. **4.** 2026 terms of \$0/t and 0 c/lb agreed by Antofagasta with a Chinese smelter, against the 2025

benchmark of \$21.25/t and 2.125 c/lb — MINING.COM, 19 December 2025. **5.** 2024 benchmark \$80/t agreed mid-November 2023; spot \$89.20/t on 4 August 2023; spot record low −\$4.10/t on 31 May 2024; 96.5% standard payability; smelter economics from by-products, sulphuric acid, gold and surplus heat — Fastmarkets analysis of negative copper concentrate treatment charges. **6.** Jiangxi Copper 2026 guidance 2.39Mt against 2.38Mt in 2025; Yunnan Copper 1.71Mt against 1.64Mt; Daye Nonferrous 713kt against 716kt; China refined copper output 14.72Mt in 2025; sulphuric acid 14.65% of Jiangxi Copper's 2025 gross profit, exceeding finished copper products; the China Smelter Purchase Team's sixteen members pledged cuts above 10% and three majors did not reduce guidance; no quarterly TC/RC guidance set for a fifth consecutive time — MINING.COM, 2 April 2026. Jiangxi first-quarter 2026 revenue RMB139.12bn (+25.39%) and net profit RMB2.82bn (+44.31%) — company announcement, April 2026. **7.** Freeport-McMoRan second-quarter 2026: revenue \$7,029m; net income \$984m (\$0.68/sh); adjusted \$1.1bn (\$0.74/sh); copper sales 710m lbs at a realised \$6.17/lb against \$4.54 a year earlier; unit net cash cost \$1.97/lb; gold sales 123koz; Grasberg guidance of about 65% of rates in the second half, 80% by mid-2027 and near-full by end-2027 — company earnings release, 23 July 2026. **8.** Grasberg mud rush of 24 September 2025: approximately 800,000t of wet material; two employees killed and five missing; force majeure declared; Goldman Sachs estimate of 250,000–260,000t of annual copper production lost across 2025–26 — The Oregon Group, September 2025. **9.** Antofagasta first-half 2026: production 285,000t (−9%); revenue \$4,479.0m (+18%); EBITDA \$2,840.5m (+27%) at a 63.4% margin; pre-tax profit \$1,995.8m (+72%); net cash costs \$1.22/lb (−8%); full-year guidance cut to 625,000–655,000t from 650,000–700,000t after weather closed Los Pelambres; net debt about \$4bn against \$2.8bn at end-2025 — company results, 13 August 2026, and MINING.COM, 13 August 2026. **10.** Aurubis nine-month FY2025/26 operating EBT €374m (+31% from €286m); third quarter €149m against €57m; full-year guidance €425–525m moved to the upper end; significantly higher sulphuric acid revenues named as a driver; spot concentrate treatment charges continuing to decline — company press release, 6 August 2026. **11.** Glencore first-half 2026: adjusted EBITDA \$10.115bn; net income \$4.405bn; marketing adjusted EBIT \$3.3bn (+142%); full-year marketing guidance \$4.7–5.0bn; own-sourced copper 397.0kt (+15% from 343.9kt); full-year copper unit cash cost guidance raised to 225.1 c/lb from 203 c/lb citing diesel, sulphuric acid and sulphur; chief executive's remarks on benchmarks versus spot — company half-year report and production report, 29 July and 5 August 2026, and earnings call transcript, 5 August 2026. **12.** Amman Mineral: smelter completion certificate signed 18 July 2026, announced 24 July 2026, designed for 900,000 t/yr of concentrate from Batu Hijau, producing cathode, gold, silver, sulphuric acid and selenium; first-quarter 2026 net sales \$808m, EBITDA \$508m (63% margin), net income \$163m, smelter averaging about 50% run rate for cathode, with no full-year cathode guidance given; next results 27 August 2026 — company press releases, 30 April and 24 July 2026. **13.** Codelco 2026 guidance about 1.34Mt cut from about 1.37Mt; the 1.7Mt medium-term target abandoned by chairman Bernardo Fontaine in a Radio Infinita interview of 28 July 2026, citing seven consecutive years of missed projections; recovery strategy due October or November 2026; Cochilco's Chilean 2026 forecast cut to 5.27Mt, −2.6% year on year — mining.com.au, 29 July 2026; MINING.COM and SMM, 13 August 2026. **14.** About 900,000t of copper redirected to US warehouses in 2026 with US inventories projected near 1.8Mt by year end; ex-US deficit of 640,000t for 2026 and a further 170,000t in 2027; Goldman Sachs revision from a 490,000t global surplus in April to a 640,000t ex-US deficit by June; COMEX–LME spread about \$400/t by early June; Commerce missing a 30 June 2026 deadline on refined copper, with a proposed 15% tariff from January 2027 rising to 30% in January 2028 — Crux Investor, 2026. **15.** Global refined copper demand +1.3% in 2025 to 27.9Mt, China +4.0% to 16.6Mt or about 59% of the world; 2025 surplus about 600kt and a 2026 surplus of 300kt on Goldman Sachs projections; fundamental fair value about \$11,500/t with a fourth-quarter 2026 forecast of \$11,200/t; demand destruction typically once price exceeds long-run marginal cost by 40% or more; fabricators reviewing aluminium substitution; scrap released by high prices — Discovery Alert, 9 July 2026, citing Goldman Sachs and JP Morgan. **16.** Jackson Hole Economic Policy Symposium, 27–29 August 2026, themed "Financial Innovation: Implications for Payments and Policy" — Federal Reserve Bank of Kansas City. Kevin Warsh took the oath of office as Chair of the Board of Governors on 22 May 2026 — Federal Reserve Board. Reporting that Chair Warsh will deliver the keynote is secondary and was not confirmed on the Kansas City Fed programme at the time of writing.

(d) Editorial choices in this automated run. Three of the run's intended steps failed and the consequences are material. *First*, the archive folder used for the thirty-day topic cooldown check could not be reached: this scheduled run executes in an isolated cloud container with no connection to the user's computer, so the "HB Wiki Learning" folder could not be listed and no filenames could be parsed. The cooldown set therefore contains only the two editions carried forward by standing instruction — AI Capex, 4 August 2026, and Gold Miners, 5 August 2026 — both of which were excluded. It is possible, and unverifiable from here, that copper or copper smelting has run within the past thirty days; if it has, this edition is a repeat and that is the reason. *Second*, the Heyokha Wiki MCP connector required authorisation that cannot be granted in an unattended session, so the entity-scoring procedure that normally selects the topic — three points for a source in the last three days, two within seven, one within twenty-one, three bonus for a title mention — could not be run, and no theme page was available to check the selection against the eligible universe. *Third*, in the absence of both, the topic was chosen on the external hot signal alone: copper's approach to a record on 11 August, driven by an Indonesian smelter failure on 8 August, made the smelting bottleneck the most newsworthy eligible subject, and it carries a live catalyst in the Jackson Hole symposium of 27–29 August. The topic name has been rendered as "Copper Smelting Bottleneck"; it could not be matched to a Wiki page title, so tomorrow's cooldown parse should be read with that in mind. Finally, several figures were dropped rather than published: year-to-date price moves for every company named, because no correctly dated source could be found; Glencore's smelting segment earnings, which are not separately disclosed; and Amman Mineral's share price, on which three sources disagreed by more than 30%.